Name: Prarabdh Srivastava
Batch: Data Analytics Dec Live Batch
Assignment: Handling Missing Data

Q1. What are the most common reasons for missing data in ETL pipelines? Answer:

The most common reasons for missing data in ETL pipelines include:

1. Source system issues ○ Data not captured at entry time ○ Optional fields left blank by users
2. Data extraction errors ○ Failed or partial data pulls
   - Network or API outages during extraction
3. Schema or format changes ○ New columns added or removed in source systems ○ Mismatch between source and target schemas
4. Data integration problems ○ Missing joins due to unmatched keys ○ Incomplete merges from multiple data sources
5. Filtering and validation rules ○ Records dropped due to strict validation checks ○ Null values removed during transformation
6. Data corruption or loss ○ File corruption
   - Truncated records during transfer or loading

These issues make it essential to include data quality checks and logging at each ETL stage.

Q2. Why is blindly deleting rows with missing values considered a bad practice in ETL?

Answer: Blindly deleting rows with missing values is considered bad practice in ETL for several reasons:

1. Loss of valuable data
   Rows may still contain useful information in other columns. Deleting them reduces data volume unnecessarily.
2. Bias in analysis
   Removing missing values can distort patterns and lead to biased or misleading results, especially if the data is not missing at random.
3. Reduced representativeness
   Certain groups or conditions may be over-represented in deleted rows, harming data quality.

---

4. Impact on downstream processes
   Fewer records can weaken machine learning models, reports, or trend analysis.
5. Loss of business context
   Missing values may indicate real-world situations (e.g., unknown age, not applicable), which should be handled meaningfully rather than discarded.

Q3. Explain the difference between: Listwise deletion Column deletion Also mention one scenario where each is appropriate.

Answer: Difference Between Listwise Deletion and Column Deletion**1. Listwise Deletion (Row Deletion)**

- **Meaning:**
  Entire rows (records) are removed if **any value is missing** in that row.
- **Effect:**
  Reduces the number of records but keeps all columns.

Appropriate scenario:

- When the dataset is **large** and the amount of missing data is **very small**, so removing a few rows does not affect analysis (e.g., survey data with <1% missing values).

---

2. Column Deletion

- **Meaning:**
  Entire columns (attributes) are removed if they contain a **high proportion of missing values**.
- **Effect:**
  Preserves rows but reduces the number of features.

Appropriate scenario:

- When a column has **too many missing values** (e.g., 70–80%) or is **not critical** for analysis, such as an optional feedback field rarely filled by users.

Q4. Why is median imputation preferred over mean imputation for skewed data such as income?

Answer: Median imputation is preferred over mean imputation for **skewed data (such as income)** because:

1. **Robust to outliers** ○ Income data usually has extreme high values. ○ The **mean is heavily influenced** by these outliers, while the **median is not**.

2. **Better representation of central tendency** ○ In skewed distributions, the median better reflects a “typical” value than the mean.
3. **Preserves data distribution** ○ Using the median helps maintain the original shape of the skewed data, whereas the mean can distort it.
4. **More stable imputation** ○ Median provides consistent replacement values even when extreme values exist.

Example:
If most incomes are ₹20,000–₹40,000 but a few are ₹5,00,000+, the mean will be unrealistically high. The median remains representative of the majority.

Q5. What is forward fill and in what type of dataset is it most useful?

Answer: Forward Fill (Forward Imputation)

Forward fill is a missing-value handling technique where a null or missing value is replaced with the most recent non-missing value that appears before it in the dataset.

How it works

If a value is missing at time *t*, forward fill copies the value from time *t-1*.

Example:

Date	Temperature
Day 1	30
Day 2	*missing*
Day 3	32

After forward fill:

Date	Temperature
Day 1	30
Day 2	30
Day 3	32

Most Useful Dataset Type

Time-series or sequential data, such as:

- Sensor readings
- Stock prices
- Weather data
- Daily sales or inventory levels
- Log and monitoring data

Why it is suitable for these datasets

- Assumes the last observed value remains valid until a new value is recorded
- Maintains temporal consistency
- Prevents sudden gaps in continuous processes

When to avoid forward fill

- When values change frequently or unpredictably
- When the missing value represents a true absence, not a delay in measurement

Q6. Why should flagging missing values be done before imputation in an ETL workflow?

Answer: Flagging missing values before imputation in an ETL workflow is important for the following reasons:

1. Preserves original information ○ Once missing values are imputed, it becomes impossible to know which values were originally missing.
   - A flag (e.g., is_missing = 1/0) keeps this information intact.
2. Improves model and analysis accuracy ○ Missingness itself can be informative (e.g., income missing more often for certain user groups).
   - Models can use the flag as an additional feature to capture this pattern.
3. Enables auditing and transparency ○ Data engineers and analysts can track how much and where imputation was applied.
   - Helps with debugging and data quality reporting.
4. Supports business interpretation ○ Imputed values are estimates, not real observations.
   - Flags let stakeholders distinguish between actual data and filled data.
5. Prevents misleading assumptions ○ Without flags, downstream users may treat imputed values as true values, leading to incorrect conclusions.

Example

Before imputation:

- Age = NULL
- Age_Missing_Flag = 1

After imputation:

- Age = 30
- Age_Missing_Flag = 1

Q7. Consider a scenario where income is missing for many customers. How can this missingness itself provide business insights?

Answer: When income is missing for many customers, the *missingness itself* can be a valuable business signal, not just a data quality issue. Here's how it can provide insights:

1. Customer Behavior and Sensitivity

- Customers may avoid sharing income due to privacy concerns or discomfort.
- Indicates high sensitivity around financial information.

Business insight:
Marketing and data-collection strategies may need better transparency or incentives.

2. Customer Segment Identification

- Missing income may correlate with specific groups:
  - Students
  - Gig workers
  - Informal-sector employees
  - High-net-worth customers who refuse disclosure

Business insight:
Income missingness can act as a proxy feature for hidden customer segments.

3. Product or Channel Issues

- Income may be missing more frequently in:
  - Certain onboarding channels (mobile app vs branch)
  - Self-service forms with optional fields
  - Legacy systems

Business insight:
Highlights gaps in form design, UX, or data capture processes.

4. Trust and Brand Perception

- High missing income rates can suggest:

- Low customer trust
- Fear of misuse of personal data

Business insight:
Signals a need for better communication on data usage and security.

5. Risk and Credit Decision Signals

- In finance, missing income often correlates with:
  - Higher uncertainty
  - Different risk profiles

Business insight:
Missing-income flags can improve credit scoring or risk models when treated as a feature.

6. Regulatory or Cultural Factors

- Cultural norms or local regulations may discourage income disclosure.

Business insight:
Helps tailor data policies by region or customer base.